Additional file 2. TRIPOD+AI checklist

Reporting checklist for the TRIPOD+AI statement (Collins et al., BMJ 2024;385:e078378). D, development; E, evaluation. Sections refer to the main manuscript.

Section	Item	D/E	Topic (shortened)	Reported in	Status
Title	1	D;E	Study type, population, outcome	Title	Reported
Abstract	2	D;E	TRIPOD+AI for Abstracts	Abstract	Reported (structured abstract; check against the abstract checklist before submission)
Background	3a	D;E	Context and rationale; existing models	Background	Reported
	3b	D;E	Target population, intended purpose and users	Background (aims); Conclusions	Reported: a methodological comparison in a population cohort; the Conclusions state that the models are not intended for clinical use
	3c	D;E	Known health inequalities	–	Not reported: sex differences are analysed (Results), but known inequalities are not discussed in the Background
Objectives	4	D;E	Objectives; development and/or validation	Background (final paragraph)	Reported
Data	5a	D;E	Data sources, rationale, representativeness	Methods: Study population; Results: Table 1	Reported (included vs excluded comparison)
	5b	D;E	Dates of data collection and follow-up	Methods: Study population	Partly: access period reported; collection years and birth cohorts are author input (M-06)
Participants	6a	D;E	Setting, centres	Methods: Study population	Reported (population-based national twin cohort)
	6b	D;E	Eligibility criteria	Methods: Study population; Fig. 1	Reported
	6c	D;E	Treatments	–	Not applicable: treatment data were not available in the analysis data and not used
Data preparation	7	D;E	Pre-processing and quality checks	Methods: Predictors (variable screening, missing codes)	Reported; similarity across sociodemographic groups not assessed
Outcome	8a	D;E	Outcome definition, timing, rationale	Methods: Outcomes	Reported (instruments, response scales and scoring)
	8b	D;E	Qualifications of outcome assessors	–	Not applicable: self-reported questionnaires
	8c	D;E	Blinding of outcome assessment	–	Not applicable: self-report
Predictors	9a	D	Choice and pre-selection of predictors	Methods: Predictors (variable screening); Additional file 1: Table S4	Reported (outcome-informed screening disclosed as a limitation)
	9b	D;E	Definition and timing of predictors	Methods: Predictors; Additional file 1: Table S4	Partly: age-9 instruments and informants are author input (M-06)
	9c	D;E	Qualifications of predictor assessors	–	Not applicable: questionnaire and genotype data
Sample size	10	D;E	How study size was arrived at, and adequacy	Methods: Study population; Discussion	Reported: set by the available complete cases, no formal calculation; the Discussion relates sample size to model complexity
Missing data	11	D;E	Handling of missing data	Methods: Study population; Fig. 1; Discussion (limitations)	Reported (complete-case analysis)
Analytical methods	12a	D	Use and partitioning of data	Methods: Data partitioning	Reported

Section	Item	D/E	Topic (shortened)	Reported in	Status
	12b	D	Handling of predictors (form, scaling)	Methods: DCMFNet, mixed-effects and machine-learning comparators	Reported
	12c	D	Model type, building steps, tuning, internal validation	Methods; Table 2; Additional file 1: Tables S1, S2b, S5	Reported (test-set involvement in DCMFNet configuration disclosed)
	12d	D;E	Heterogeneity across clusters	Methods: Data partitioning, mixed-effects comparators	Reported (twin pairs as clusters: grouped splits; random intercept in the mixed models)
	12e	D;E	Performance measures and model comparison	Methods: Evaluation and statistical comparison; Calibration, subgroup and sensitivity analyses	Reported
	12f	E	Model updating	–	Not applicable: no recalibration or updating
	12g	E	How predictions were calculated	Methods: Software and reporting	Reported: code archived with a DOI
Class imbalance	13	D;E	Class-imbalance methods	Methods: Loss	Reported (continuous outcomes; tail-weighted loss in DCMFNet)
Fairness	14	D;E	Approaches to fairness	Methods: Calibration, subgroup and sensitivity analyses	Partly: performance by sex reported; no fairness intervention
Model output	15	D	Output and any thresholds	Methods: Outcomes	Reported (continuous scores, no thresholds)
Training vs evaluation	16	D;E	Differences between development and evaluation data	Methods: Data partitioning	Reported (random splits of one cohort; no differences by design)
Ethical approval	17	D;E	Ethics committee and consent	Methods: Ethics; Declarations	Author input (C-02)
Funding	18a	D;E	Funding and role of funders	Declarations	Author input (C-03)
Conflicts of interest	18b	D;E	Competing interests	Declarations	Author input (C-03)
Protocol	18c	D;E	Protocol access	Methods: Software and reporting	Reported: no protocol was published
Registration	18d	D;E	Study registration	Methods: Software and reporting	Reported: not registered
Data sharing	18e	D;E	Data availability	Declarations: Availability of data and materials	Partly: access procedure is author input (C-07)
Code sharing	18f	D;E	Code availability	Methods: Software and reporting; Declarations	Reported: Zenodo DOI and GitHub repository
Patient and public involvement	19	D;E	Patient and public involvement	Methods: Software and reporting	Reported: no involvement
Participants	20a	D;E	Participant flow	Results: Analysis sample; Fig. 1	Reported
	20b	D;E	Characteristics and missing data	Results: Table 1	Reported
	20c	E	Comparison with development data	–	Not applicable: internal evaluation on random splits of the same cohort
Model development	21	D;E	Numbers in each analysis	Methods: Data partitioning; Fig. 1	Reported (training, validation and test sizes)
Model specification	22	D	Full model available for new predictions	Methods; Table 2; Additional file 1: Table S3	Partly: mixed-model coefficients and all code given; network weights trained on CATSS are not shared

Section	Item	D/E	Topic (shortened)	Reported in	Status
Model performance	23a	D;E	Performance with confidence intervals, incl. subgroups	Results: Tables 3–5	Reported: mean and SD over splits for each model; paired differences with 95% confidence intervals and corrected p values (Table 4); results by sex (Table 5)
	23b	D;E	Heterogeneity across clusters	–	Not applicable: twin pairs are too small for cluster-level performance
Model updating	24	E	Results of updating	–	Not applicable
Interpretation	25	D;E	Overall interpretation, incl. fairness	Discussion	Reported
Limitations	26	D;E	Limitations and their effects	Discussion: Strengths and limitations	Reported
Usability	27a	D	Handling poor or missing input data in use	–	Not applicable: no model is proposed for use
	27b	D	User interaction and expertise	–	Not applicable: no model is proposed for use
	27c	D;E	Next steps for research	Discussion; Conclusions	Reported